Precogs: Humans with the ability to predict a crime even before it is committed. Thus went the premise of the 2002 hit movie, *Minority Report*. Can science ever develop a piece of software or hardware to predict the future? ?

Humans bring with them cognitive biases: they rely too much on one set of information without looking at the entire picture formed, often with disastrous results. To minimise errors and to eventually produce a silicon stock market capable of simulating the lows and ups of the real thing, AI techniques are pitched together. Rule sets are created with data extracted from market snapshots of the past. Software agents are then hired to simulate buying and selling decisions based on these rules. These decisions are then passed on to a human who does the actual transactions. Each decision made by such an agent is accorded a weightage based on the degree of success of the choice. Successful agents are then 'bred' together—a snippet of code from here, a snip from there and you have birthed something new. Mutations can also be introduced in the progeny by toggling the 1s and the 0s involved. The resultant offspring are similarly evaluated on their successes and the process is repeated till a healthy ecology emerges wherein software agents of repute move up the ladder while the faulty ones fade into extinction, taking the associative rule sets with them. To add to the heady mix are fuzzy logic elements that can base decisions on a range of data, rather than discrete values of the same. Such cocktails have been used to construct expert systems—software powerhouses designed to replace a human professional such as a market analyst.

Built upon the power of supercomputers, an expert system can perhaps one day answer the age-old questions about 'Life'. Nirvana, via your friendly neighbourhood computer!

Bicentennial Man, the 1999 film starring Robin Williams told of Andrew Martin, a household android who begins to feel genuine human emotions. Will a robot ever reach a level of intelligence to be considered sentient and maybe, even human? ?

Move over AIBO, enter Robokoneko. Proving comprehensively that cats are indeed smarter than dogs, the robot kitten is all set to take centre stage with a brain powered by the electrical firings between some 40 million artificial neurons. The brain will consist of 72 programmable silicon chips whose connections between constituent transistors can be electronically rewired. To maximise their potential, connections will repeatedly be reconfigured to represent 32,768 different modules. As the brain cycles through these modules at a rate of 300 times per second, the effect will be akin to a singular entity with a virtual bank of 37.7 million neurons. Since the task of programming a bank as complex as the kitty's brain seems illogical, the scientists turn to genetic algorithms for respite. Codes of software will be let to brew, intermingle and share snippets amongst each other, eventually leading to the creation of a neural network based on the best-of-breed software snippets.

Take a brain as complex as Robokoneko's, add in the expressions and the emotional ability of a Kismet and stir vigorously. Blend in face recognition software, speech analyzers, speech synthesizers, a healthy dollop of automated limbs and voila, a fully functional android is at your service. Happy living.

"Agent Smith", say it in the soporific drone of the villain that plagued Neo and the crew in *The Matrix*. Smith was created as a figure to police maintain hegemony the humans in the vi network that enmesh the fictional world. C scientists come up w software codes that serve the Internet in similar fashion?

How are viruses, bacteria a Internet inter-related? Se Net is a mass of connections, together at nodes that phys manifest as routers. These r have a fixed buffer capacity determines their ability to a store and re-route data. Some the data arriving at these noc lost by the router. What is n then, is a mechanism by whic buffers monitor themselves, a the incoming packets and their lengths accordingly. So does sex come in? Bacteria duce in a very efficient, albei exciting manner: when two cells come in contact with other, genes snippets are excha Successful genes are propa while ineffective ones fade Software codes behave simil strings of data imitate the gen pets that are then collated anc spersed between the software. code snips let the buffer adap evolve as per the needs of the

This could be put to use phone lines, making droppe and busy lines a thing of the